ONENET SOLUTIONS · MIGRATION GUIDE

HR & Payroll Suite Data Migration from CyberAP

For your IT administrator: move employees, 12+ years of attendance and the payroll history from your old CyberAP (SQL Server) system into your new HR & Payroll Suite yourself, without OneNet having to touch your data.

1 · Prepare
Backup, permission, cut-over date

2 · Export
PowerShell on the SQL Server PC

3 · Import
Your HR system → Data Import

4 · Go live
Checks, HR clean-up, gap import

Source
CyberAP on SQL Server 2008–2022

Target
PostgreSQL, Full or Lite edition

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What is migrated, archived and left behind

CYBERAP	NEW SYSTEM	NOTES
branch, shift	Branches; shift on each employee	Province is set to Punjab and city is blank: correct them in Company Setup.
employee_group	Departments	Sub-group is kept on the employee (legacy details).
designation, holiday	Designations, Holidays	Branch off days are archived.
employee (all, active and former)	Employees	Code, name, father's name, CNIC, phones, addresses, dates, gender, marital status, branch, department, designation, shift, photo. Former staff become *Terminated*.
salary (basic salary)	Basic salary	Only when it is a real figure; CyberAP's placeholder 1.0 becomes 0.
attendance, processed_attendance, employee_over_time, employee_tracking, employee_time_out_reason	Attendance (one record per employee per day)	Status, duty in/out, every break with its reason (Prayer, Lunch, Personal …), overtime, branch, personal-out minutes. CyberAP codes are kept on the record.
Payroll 2015–2018, monthly summaries, salary heads, advances, users, settings, screens	Archive table `legacy_records`	Kept for reference and audits; not shown in payroll screens.
Fingerprint templates, original photos	Archive table `legacy_records`	Fingerprints are DigitalPersona One Touch templates; the kiosk needs U.are.U ANSI templates, so staff are enrolled again.
User passwords, branch connection strings, sa passwords	**Never copied**	Removed by the exporter itself.

Before you start

- [ ] **CyberAP backup:** a fresh `.bak` of the CyberAP database from the head-office server, or access to that server.

- [ ] **Branch databases:** confirm that every branch (`CyberAP-AT`) has synced its punches to head office. The export reads head office only.
- [ ] **Windows PC** with SQL Server (Express is enough) and Windows PowerShell 5.1. Nothing else needs to be installed.
- [ ] **Data Import switched on** for your installation by your service provider (OneNet). When it is on, Super Admins see **Data Import** in the menu under SYSTEM.
- [ ] **Permission:** a Super Admin, or a user whose role has the **Data import** permission (User Management → Roles).
- [ ] **Your password at hand:** removing current data asks for it again.
- [ ] **Large package?** Packages of a few hundred MB upload fine but take time on a slow line; your service provider can also import from the server (see that section).
- [ ] **Cut-over date** agreed with HR: the day the kiosks replace CyberAP.

> **Analyse first, then import**
> Analysis reads the whole package and shows what would happen without changing anything. Review its report with HR before importing. An import can be re-run safely: nothing is duplicated.

Step 1 — Restore the CyberAP backup (if needed)

Skip this step when you export directly from the head-office SQL Server. Otherwise restore the `.bak` on the Windows PC with SQL Server Management Studio (Databases → Restore Database → Device) or with:

```
sqlcmd -S .\SQLEXPRESS -E -Q "RESTORE FILELISTONLY FROM DISK='D:\CyberAP.bak'"
sqlcmd -S .\SQLEXPRESS -E -Q "RESTORE DATABASE CyberAP FROM DISK='D:\CyberAP.bak' WITH MOVE 'CyberAP' TO
'D:\SQL\CyberAP.mdf', MOVE 'CyberAP_log' TO 'D:\SQL\CyberAP_log.ldf', RECOVERY"
```

Use the logical names that RESTORE FILELISTONLY prints. The export only reads the database; it never changes it.

Step 2 — Export CyberAP on Windows

> **HR & Payroll installed on Windows Server? Skip the script.**
> On a Windows Server installation the Data Import page has **Create package from SQL Server**: enter the SQL Server instance (it is suggested), sign in with a SQL login, press **Connect and list databases**, pick the database marked *CyberAP* and press **Create package**. The server runs this same exporter itself and the package is ready for Analyse when it finishes; nothing has to be copied or uploaded. The SQL password is used only for that export and is never stored. See the *Windows Server Guide* for a read-only SQL login.

Copy `Export-CyberAP.ps1` (in the repository under `tools\cyberap-export\`, and next to this PDF in `deploy\installer\dist\`) to the PC and run it in PowerShell:

```
# local SQL Server, signed in with your Windows account
powershell -ExecutionPolicy Bypass -File .\Export-CyberAP.ps1

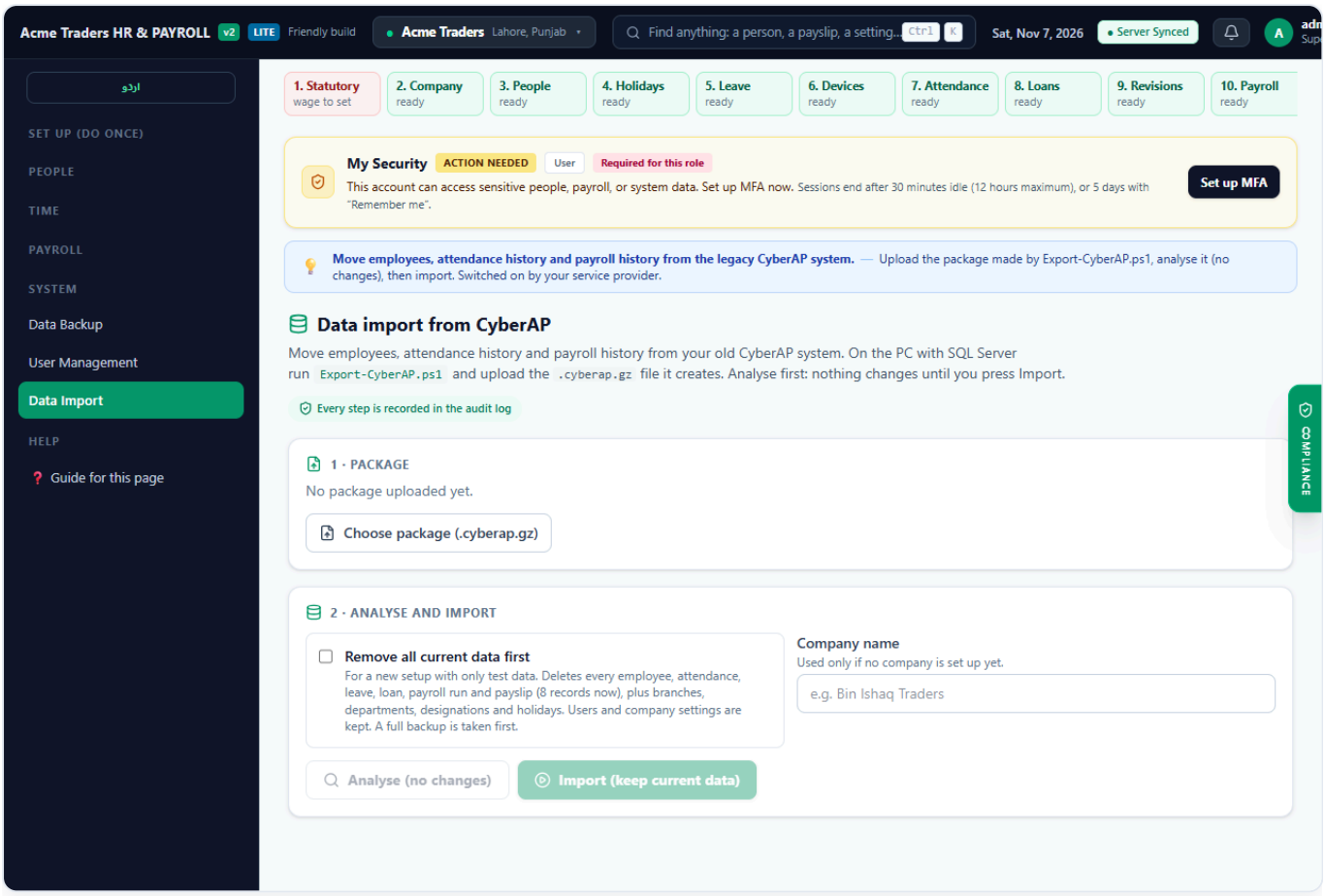
# another server or a SQL login
powershell -ExecutionPolicy Bypass -File .\Export-CyberAP.ps1 -Server 192.168.86.83 -Database cyberap -
SqlUser sa -SqlPassword "…" -OutFile D:\cyberap.cyberap.gz
```

PARAMETER	DEFAULT	MEANING
`-Server`	`.\SQLEXPRESS`	SQL Server instance.
`-Database`	`CyberAP`	Database name.
`-SqlUser / -SqlPassword`	Windows login	Use a SQL login instead of Windows authentication.
`-OutFile`	`CyberAP-<date>.cyberap.gz`	Package file to write.
`-FromDate YYYY-MM-DD`	all history	Attendance from this date only (employees and masters are always included). Used for the gap import on cut-over day.
`-NoPhotos`	off	Skip photos (smaller and faster; for test runs).
`-ThumbSize / -PhotoSize`	160 / 480	Pixel size of the profile thumbnail and of the archived photo. Originals (up to 36 MB each) are never copied.

The script prints its progress (tables, employees, attendance days every 100,000) and ends with the package path and size. Keep the window open until it says **Package written**; a cut-off package is rejected by the import.

Step 3 — Open Data Import in your HR system

1. Sign in to your HR system as a Super Admin (or a user with the **Data import** permission).
2. Menu → SYSTEM → **Data Import**. If the item is missing, ask your service provider to switch data import on.



Data Import in your HR system: upload the package, analyse, then import. Every step is recorded in the audit log.

Step 4 — Upload and analyse

1. **Choose package (.cyberap.gz)** and select the file. A progress bar shows the upload; a broken connection is retried and resumes where it stopped.
2. When the upload ends the package is checked and shown with its source server, database and export time.
3. Press **Analyse (no changes)**. The whole package is read and a report appears: employees, attendance per status and period, masters, archive rows and a **package check** (row counts against the export). Nothing is written.
4. Read the warnings with HR (see Step 7). Typical ones: active staff with no recent attendance, missing CNICs, placeholder salaries, fingerprints to enrol again.

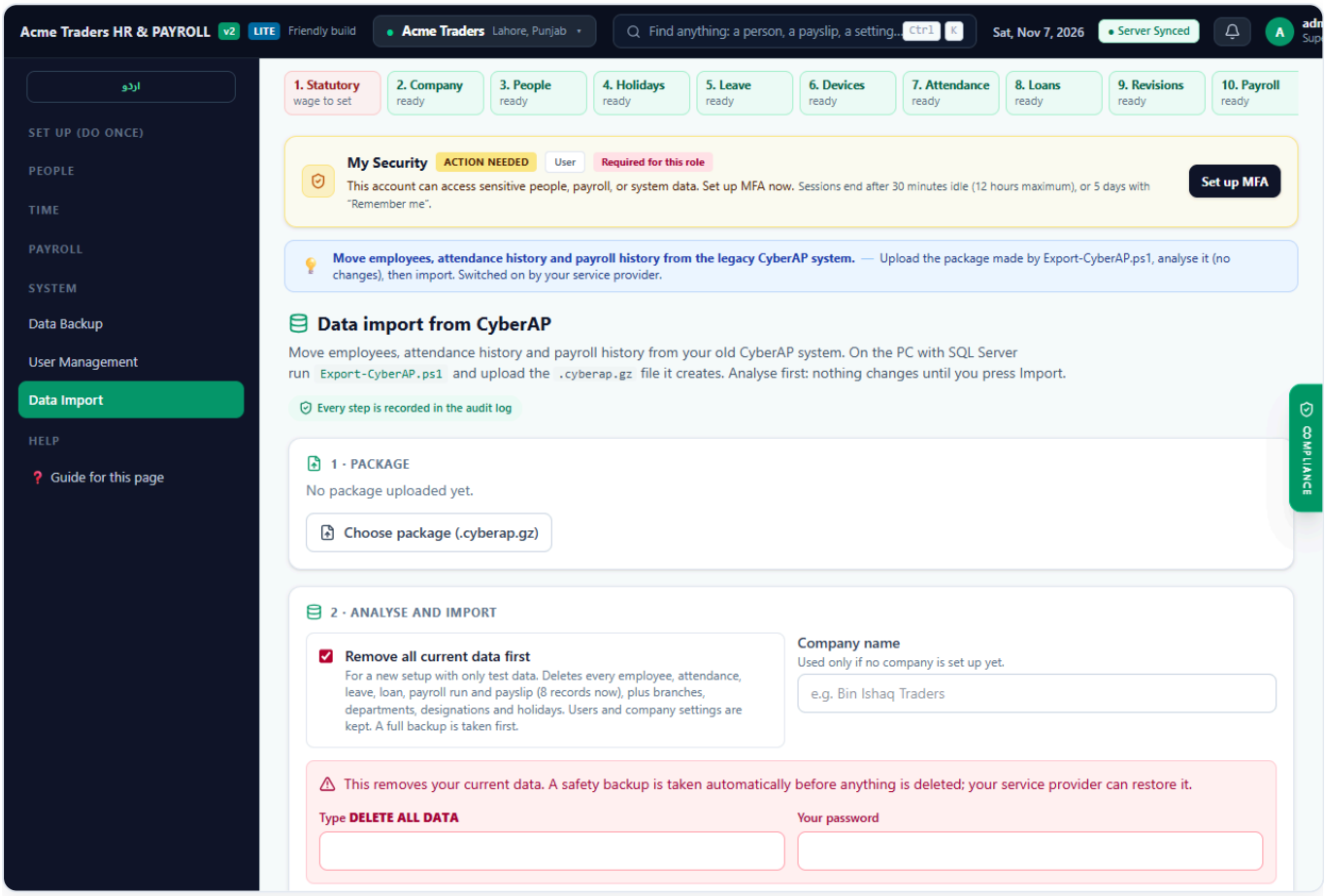
Step 5 — Import (keep or remove current data)

Keep current data

Leave *Remove all current data first* unticked and press **Import (keep current data)**. Legacy employees whose code already exists are linked to the existing employee instead of being created twice, and an attendance day is never written over a day the new system already recorded.

Remove current data (new setup)

Tick **Remove all current data first**, type `DELETE ALL DATA`, enter **your password** and press **Back up, remove current data and import**. Use this only when the system holds test or demo data.



Removing current data needs the typed confirmation and your password. A full backup is taken automatically first.

SAFETY RAIL	WHAT IT DOES
Automatic safety backup	Before anything is deleted, the system takes a full encrypted backup of its database. If the backup cannot be made, nothing is deleted. Your service provider can restore it.
Password and confirmation	The person importing re-enters their password and types `DELETE ALL DATA`.
Locked after go-live	Once salaries have been paid or an attendance month has been locked, removing data is refused for good; only "keep current data" imports remain possible.
Audit	Who imported what, and when, is written to the audit log.

REMOVED BY "REMOVE ALL CURRENT DATA"	ALWAYS KEPT

`employees, biometric_templates, attendances, attendance_periods, mobile_duty_authorizations, leaves, holidays, payroll_runs, payroll_payslips, payroll_adjustments, loan_advances, salary_revisions, gratuity_settlements, performance_reviews, company_assets, job_postings, job_applications, notifications, lifecycle_requests, employee_documents, payslip_deliveries, payslip_alert_deliveries, branches, departments, designations, zones, uc_towns, hr_documents, legacy_records`	companies, users, roles, auth sessions and MFA, wage_types, tax_slabs, stat_config, statutory_change_requests, kiosk_devices, app_settings (edition), hr_audit_log, schema_migrations. Users stay and can still sign in; their link to a deleted employee is cleared.

The import runs on the server in the background. The screen shows the phase (safety backup, removing data, archiving, employees, attendance history, finishing) and a progress bar. Closing the browser does not stop it; open Data Import again to see progress. **Cancel** stops after the current batch; what was written stays and a re-run continues.

> **Removing data cannot be undone by you**
> Only your service provider can restore the automatic safety backup. Use "Remove all current data" only for a new setup.

For your service provider — server import

For very large packages or slow connections, the provider can copy the package to the server (aaPanel → Files → Upload, or `scp`) and run the importer in the app folder (aaPanel → Terminal). It reads `DATABASE_URL` from the app's `.env`:

```
cd /www/wwwroot/<hr-app>
node scripts/cyberap-import.cjs --package /root/cyberap.cyberap.gz --analyze
node scripts/cyberap-import.cjs --package /root/cyberap.cyberap.gz
node scripts/cyberap-import.cjs --package /root/cyberap.cyberap.gz --wipe --confirm "DELETE ALL DATA" --
company-name "Client name"
```

Progress is printed every few seconds and the full report at the end. Restart the app afterwards in aaPanel's Node manager so the new attendance window is picked up immediately (it is otherwise read within 30 seconds).

Step 6 — Check the result

- [ ] Data Import says **Import · finished** and the package check is **ok**.
- [ ] Sign in as the Super Admin. Employees shows the active staff with photos; former staff appear as Terminated.
- [ ] Attendance shows the current and previous month with in/out times and breaks; an employee's profile shows their full history.
- [ ] Company Setup lists the CyberAP branches, departments and designations.
- [ ] Optional, for your service provider: SQL check of the database:

```
SELECT status, count(*) FROM employees GROUP BY status;
SELECT date_trunc('year', date) AS year, count(*) FROM attendances GROUP BY 1 ORDER BY 1;
SELECT kind, count(*) FROM legacy_records GROUP BY kind ORDER BY 2 DESC;
SELECT id, status, started_at, finished_at FROM legacy_import_runs ORDER BY started_at DESC;
```

Step 7 — HR clean-up before the first payroll

ITEM	WHAT TO DO
Basic salary	CyberAP holds 1.0 for most staff. Enter each active employee's real salary (Employees → edit, or Salary Revisions). Payroll refuses or underpays a 0 salary.
Active staff with no recent attendance	The report lists employees who are active in CyberAP but have not worked for 30+ days. Mark those who left as Terminated with their separation date.
CNIC	Placeholders such as "....." are imported as blank; the original text is kept in the legacy details. Enter real CNICs (Full edition: required for EOBI/FBR).
Fingerprints	Enrol each active employee again at the kiosk (U.are.U reader). Face recognition can be enrolled at the same time.
Bank, EOBI, NTN, email	Not in CyberAP. Full edition: complete them before the first payroll. Lite edition: optional.
Branches	Set city, province, address and geofence for each branch.
Attendance policy	CyberAP used: shift 11:00–23:30, 15 minutes grace, date change at 06:00, minimum overtime 10 minutes, "Duty Off" allowance 90 minutes, holiday sandwich rule on. Set the matching options in the attendance policy.
Holidays	CyberAP's holiday list ends in 2015. Add the current year's holidays.

Step 8 — Cut-over day: the gap import

CyberAP keeps recording punches until the kiosks take over. On cut-over day bring over the days since the main export:

1. Stop using CyberAP terminals (or note the time).
2. Export only the recent days: `.\Export-CyberAP.ps1 -FromDate 2026-11-01 -NoPhotos` (use the first day not fully covered by the main package).
3. Upload and import it **without** removing data. Existing employees and edited records are left as they are; new employees are added; days already recorded by the kiosk are kept.
4. Switch the branches to the kiosks.

How imported history is shown

Browsers load attendance as one list. After an import of more than 10,000 days the server turns on an attendance window (setting `dataLimits`): manager screens receive the **current and previous month**, newest first, and an employee's own history is not limited. The whole history stays in the database. To show more months:

```
UPDATE app_settings SET data = data || '{"attendanceReadMonths": 3}' WHERE key = 'dataLimits';
```

> **Large staff numbers**
> With 500+ active employees, every manager browser reloads the current month's attendance every few seconds. Keep the window at 1 month, enable gzip for JSON in nginx (aaPanel → Website → Config, `gzip_types application/json`) and give the server enough bandwidth.

Troubleshooting

MESSAGE	CAUSE AND FIX
Cannot connect to … (export)	Wrong `-Server` name, SQL Server service stopped, or the Windows account has no access. Try `sqlcmd -S .\SQLEXPRESS -E -Q "SELECT 1"`.
This file is not a CyberAP export package	Upload the `.cyberap.gz` made by Export-CyberAP.ps1, not a `.bak` or a renamed file.
Package check: failed / no end marker	The export was interrupted or the file was cut off in transfer. Export again and re-upload.
Data Import is not in the menu / "switched off for this installation"	Ask your service provider to switch client data import on.
"You need the Data import permission"	A Super Admin adds the Data import permission to your role (User Management).
"Backups are not configured … cannot be removed safely"	The server has no backup key. Ask your service provider; meanwhile you can import with "keep current data".
"This installation is live … can no longer be removed"	Payroll was paid or a month locked. Import with "keep current data" (cut-over gap imports work this way).
"Your password is not correct"	Enter the password of the account you are signed in with.
HTTP 413 during upload	The server's upload limit is below 8 MB. Ask your service provider to raise nginx `client_max_body_size`.
Import failed: no space left on device	Ask your service provider to free disk space, then upload again.
Server restarted during import	The status shows the last saved phase. Run the import again (without removing data): it continues and skips what is already there.
Kiosk says enrolled templates are from another reader	A legacy One Touch template was entered manually. Delete it and enrol the employee again on the kiosk.

Rollback

Ask your service provider to restore the automatic safety backup taken just before the import (its time is shown on the Data Import screen). This is the only complete rollback; there is no "undo import" button, because an import that removed data cannot bring it back.

Appendix A — Field and status mapping

Day status

CYBERAP PROCESSED DAY (REMARKS1 / REMARKS2 / REMARKS4)	NEW STATUS
A, or "-"	Absent — **unless the employee punched Duty In**: CyberAP pre-fills days as A until they are processed, so such days become Present or Late from the punches (marked *correctedFromAbsent*).
H (holiday), BD (branch off day)	Holiday
P with remarks4 = HD	Half Day
P with remarks2 = LI (late in)	Late
P otherwise, ED (extra day)	Present
No processed row yet (e.g. today)	Present or Late from the first Duty In and the shift start + grace

Absent days after the last working day of a former employee are dropped (CyberAP keeps generating them). The original codes, worked/out minutes, overtime approvals and per-reason out minutes are kept on each record under `legacy`.

Punches

CYBERAP PUNCH	NEW RECORD
In · Duty (first)	Punch in
Out · Duty Off (last)	Punch out
Out · Prayer / Lunch / Personal / Tea/Smoking / Office Work / Other, then In	A break with reason, out time and return time. An out without a return is an open break.
employee_tracking branch	Terminal branch; "cross branch" when it differs from the employee's branch
employee_over_time.total_over_time	Overtime minutes

Employees

CYBERAP	NEW FIELD
employee_code, employee_name	Employee code, full name
is_active = 1 and employment_status = Employed	Status Active; otherwise Terminated with separation date = leaving date
employee_cnic_number	CNIC as 12345-1234567-1 (blank when not 13 digits)
mobile, else telephone	Contact number
branch, sub-group → group, designation, shift	Branch, department, designation, shift

employee_image	160 px profile photo (active staff); 480 px copy archived for everyone
father name, bar code, addresses, blood group, religion, notes, salary heads	Legacy details on the employee record

Appendix B — Trial migration figures

The full CyberAP database (CyberAP head-office copy, SQL Server Express, 55 tables, 13.6 million rows) was exported and imported into a test installation on 7 November 2026. Use these figures to judge the time and size of a real migration.

Export (Windows PC)

Source size	4.6 GB
Export time	8 min (most of it resizing 3,952 photos)
Package size	199 MB
Attendance days in package	1,698,197
Upload (Data Import screen)	about 3 min at 10 Mbit/s (8 MB pieces)
Analysis time	about 1.5 min
Import time (with wipe)	4 min
Database size after import	1.6 GB (attendance 1.5 GB)

Imported

Employees (active / left)	4,083 (767 / 3,316)
Branches / departments / designations / holidays	9 / 29 / 38 / 12
Attendance days	1,536,319
Period	2010-08-16 → 2026-11-07
Late	307,156
Absent	446,981
Holiday	2,360
Present	777,491
Half Day	2,331
Absent days corrected from punches	6,197
Absences after leaving (dropped)	161,691
Archived legacy rows	19,626

Warnings raised for this database

- 307 employees are active in CyberAP but have no working day since 2026-10-08. They are imported as Active; review the list and mark the ones who left as Terminated.
- 189 active employees have no valid CNIC in CyberAP; they are imported with the CNIC blank.
- 690 active employees have no real basic salary in CyberAP (1.0 placeholder); they are imported with salary 0 and must be updated before the first payroll.
- 3938 fingerprints are DigitalPersona One Touch templates. The kiosk reads U.are.U ANSI templates, so they are archived and staff must be enrolled again at the kiosk.